

Decision Tree

What Is a Decision Tree?

A Decision Tree is a supervised machine learning algorithm used for both classification and regression. It works by splitting data step by step based on feature values, forming a tree-like structure that resembles human decision-making logic.

Tree-like structure in which an internal node represents a test on an attribute, each branch represents outcome of the test and each leaf node represents class label.

The model learns a series of **if-else rules** from data to arrive at a prediction. Because of this structure, decision trees are one of the **most intuitive and interpretable ML models**, especially for beginners.

Decision tree is a decision support recursive partitioning structure that uses a tree like model of decisions and their possible consequences including chance event outcomes

Rules based algorithm

Divide and conquer algorithm

Greedy algorithm

Decisions are mainly based on some conditions

Decisions made can be easily explained

Terminologies in decision tree

Root node: it represents the entire population or sample and its further gets divided into two or more homogeneous sets.

Leaf node: node cannot be further segregated into further nodes.

Splitting: dividing the root node/ sub node into different paths on the basis of some condition.

Branch /sub tree: formed by splitting the tree/node.

Pruning: opposite to splitting, basically removing unwanted branches from tree.

Parent /child node: root node is the parent node and all the other nodes branched from it are known as child nodes.

Entropy : defines a randomness in the data; it is a metric which measures the impurity. The first step to solve the problem of decision tree.

$$Entropy(S) = \sum_{i=1}^c -p_i \log_2 p_i$$

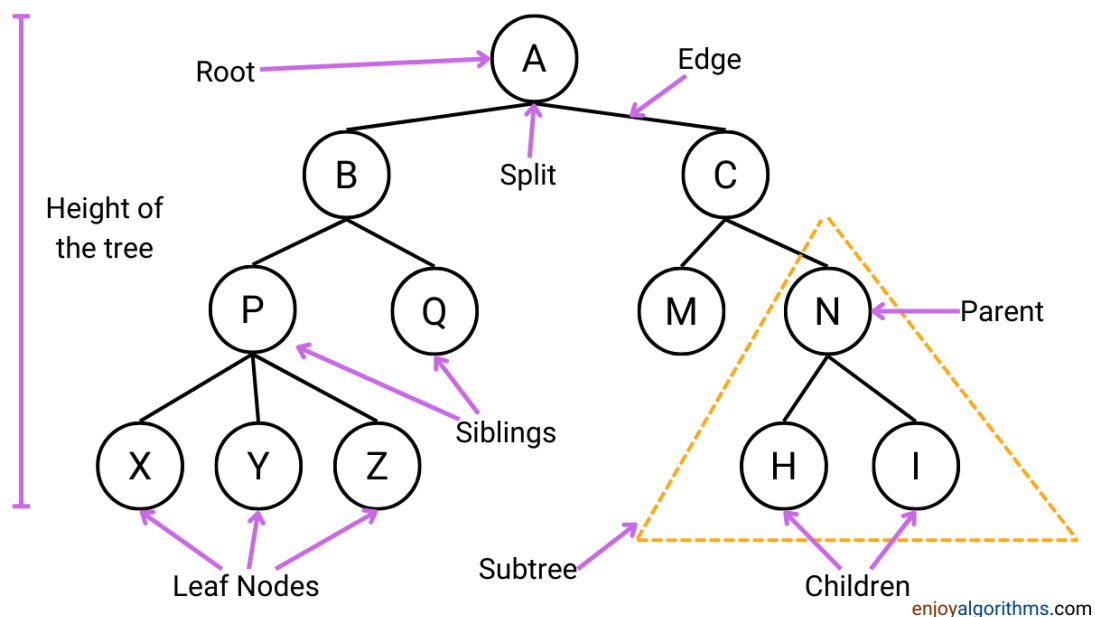
where c is the number of output classes.

Information gain: ig is the decrease in entropy after a dataset is split on the basis of an attribute.

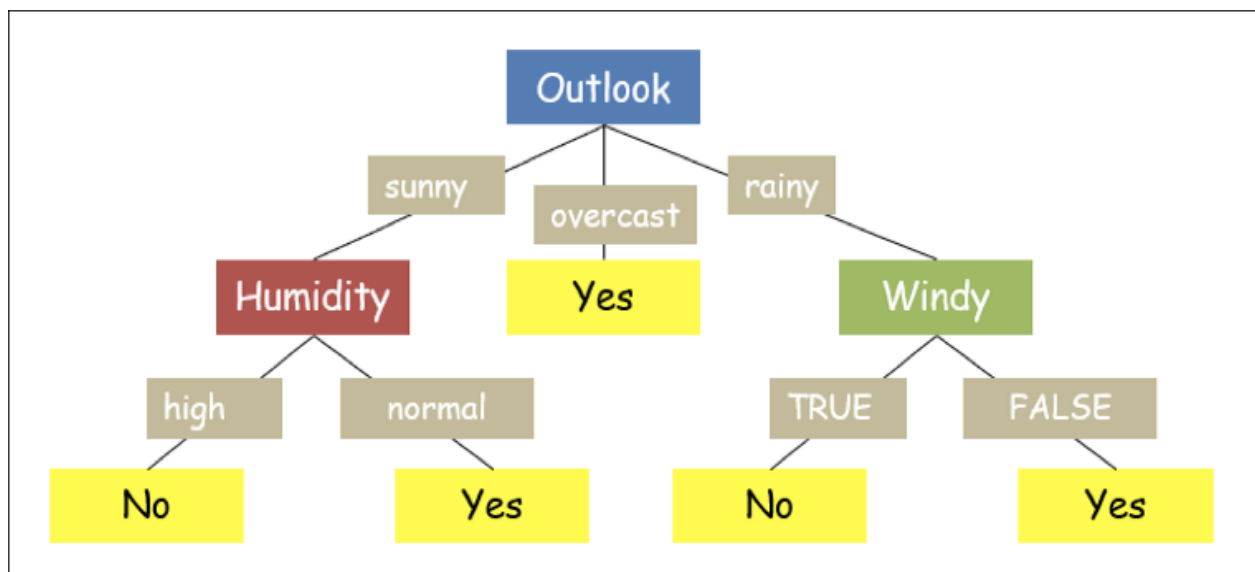
Constructing a decision tree is all about finding attributes that return the highest information gain.

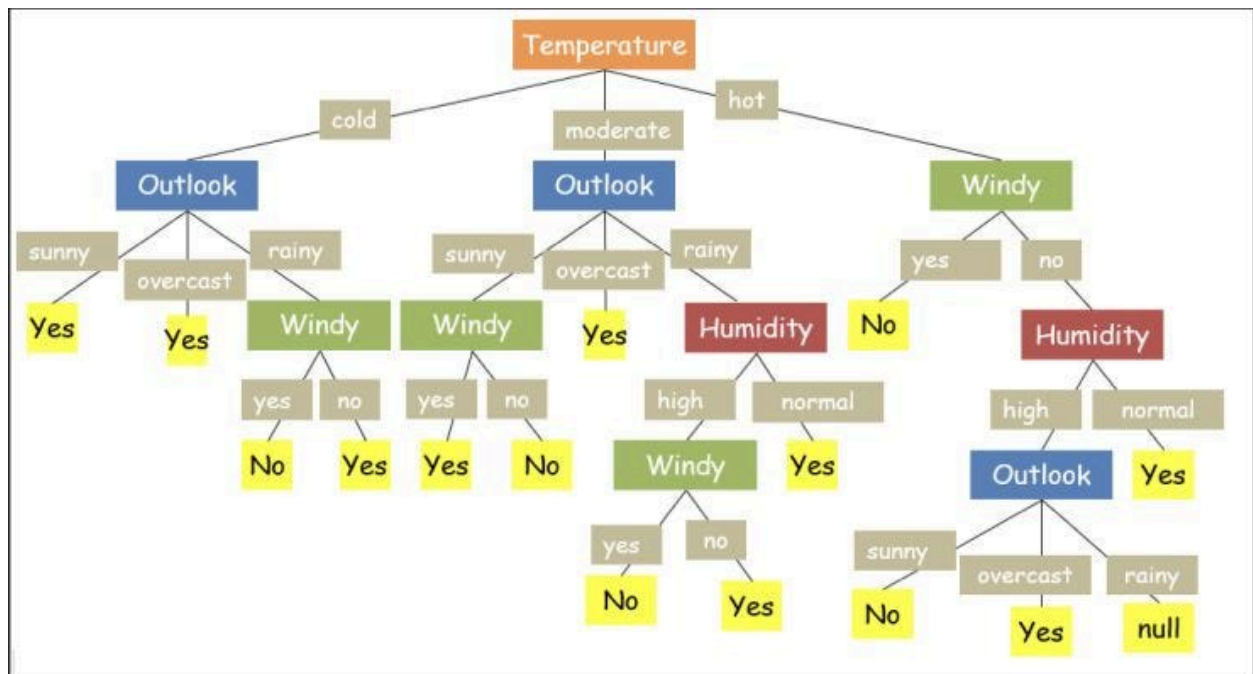
$$IG(S, A) = Entropy(S) - \sum_{v=Values(A)} \frac{|S_v|}{|S|} Entropy(S_v)$$

Gini index: the measure of impurity used in building decision tree.



Day	outlook	temp	humidity	windy	play
1	sunny	hot	high	FALSE	no
2	sunny	hot	high	TRUE	no
3	overcast	hot	high	FALSE	yes
4	rainy	mild	high	FALSE	yes
5	rainy	cool	normal	FALSE	yes
6	rainy	cool	normal	TRUE	no
7	overcast	cool	normal	TRUE	yes
8	sunny	mild	high	FALSE	no
9	sunny	cool	normal	FALSE	yes
10	rainy	mild	normal	FALSE	yes
11	sunny	mild	normal	TRUE	yes
12	overcast	mild	high	TRUE	yes
13	overcast	hot	normal	FALSE	yes
14	rainy	mild	high	TRUE	no





14 samples of the dataset, 9 are 'yes' and 5 are 'no'. Thus:

$$E(S) = - \left(\frac{9}{14} \right) \log_2 \frac{9}{14} - \left(\frac{5}{14} \right) \log_2 \frac{5}{14} = 0.94$$

Let's first calculate the IG of the "Outlook" attribute.

$$= 0.94 - \left(\frac{5}{14} \right) E(S_{\text{sunny}}) - \left(\frac{4}{14} \right) E(S_{\text{overcast}}) - \left(\frac{5}{14} \right) E(S_{\text{rainy}})$$

Day	outlook	temp	humid	windy	play
1	sunny	hot	high	FALSE	no
2	sunny	hot	high	TRUE	no
8	sunny	mild	high	FALSE	no
9	sunny	cool	normal	FALSE	yes
11	sunny	mild	normal	TRUE	yes

Day	outlook	temp	humid	windy	play
3	overcast	hot	high	FALSE	yes
7	overcast	cool	normal	TRUE	yes
12	overcast	mild	high	FALSE	yes
13	overcast	hot	normal	FALSE	yes

Day	outlook	temp	humid	windy	play
4	rainy	mild	high	FALSE	yes
5	rainy	cool	normal	FALSE	yes
6	rainy	cool	normal	TRUE	no
10	rainy	mild	normal	FALSE	yes
14	rainy	mild	high	TRUE	no

$$E(S_{\text{sunny}}) = -\left(\frac{3}{5}\right)\log_2\frac{3}{5} - \left(\frac{2}{5}\right)\log_2\frac{2}{5} = 0.97$$

$$E(S_{\text{overcast}}) = -\left(\frac{4}{4}\right)\log_2\frac{4}{4} - \left(\frac{0}{4}\right)\log_2\frac{0}{4} = 0$$

$$E(S_{\text{rainy}}) = -\left(\frac{3}{5}\right)\log_2\frac{3}{5} - \left(\frac{2}{5}\right)\log_2\frac{2}{5} = 0.97$$

Thus,

$$IG(S, A = \text{'outlook'}) = 0.94 - \left(\frac{5}{14}\right)*0.97 - \left(\frac{4}{14}\right)*0 - \left(\frac{5}{14}\right)*0.971 = 0.246$$

Similarly

$$\begin{aligned} IG(S, A = \text{'humidity'}) &= E(S) - \sum_{v=[\text{high}, \text{normal}]} \frac{|S_v|}{|S|} Entropy(S_v) \\ &= 0.94 - \left(\frac{7}{14}\right)E(S_{\text{high}}) - \left(\frac{7}{14}\right)E(S_{\text{normal}}) \end{aligned}$$

Day	outlook	temp	humid	windy	play
1	sunny	hot	high	FALSE	no
2	sunny	hot	high	TRUE	no
3	overcast	hot	high	FALSE	yes
4	rainy	mild	high	FALSE	yes
8	sunny	mild	high	FALSE	no
12	overcast	mild	high	FALSE	yes
14	rainy	mild	high	TRUE	no

Day	outlook	temp	humid	windy	play
5	rainy	cool	normal	FALSE	yes
6	rainy	cool	normal	TRUE	no
7	overcast	cool	normal	TRUE	yes
9	sunny	cool	normal	FALSE	yes
10	Rainy	mild	normal	FALSE	yes
11	sunny	mild	normal	TRUE	yes
13	overcast	hot	normal	FALSE	yes

$$E(S_{\text{high}}) = -\left(\frac{3}{7}\right)\log_2\frac{3}{7} - \left(\frac{4}{7}\right)\log_2\frac{4}{7} = 0.985$$

$$E(S_{\text{normal}}) = -\left(\frac{1}{7}\right)\log_2\frac{1}{7} - \left(\frac{6}{7}\right)\log_2\frac{6}{7} = 0.591$$

$$IG(S, A = \text{'humidity'}) = 0.94 - \left(\frac{7}{14}\right)*0.985 - \left(\frac{7}{14}\right)*0.591 = 0.152$$

Same like above we calculate for windy and temp

$$\begin{aligned}
 IG(S, A = 'windy') &= E(S) - \sum_{v=[True, False]} \frac{|S_v|}{|S|} Entropy(S_v) \\
 &= 0.94 - \frac{8}{14} E(S_{False}) - \frac{6}{14} E(S_{True})
 \end{aligned}$$

$$\begin{aligned}
 IG(S, A = 'temp') &= E(S) - \sum_{v=[hot, mild, cool]} \frac{|S_v|}{|S|} Entropy(S_v) \\
 &= 0.94 - \frac{4}{14} E(S_{hot}) - \frac{6}{14} E(S_{mild}) - \frac{4}{14} E(S_{cool})
 \end{aligned}$$

By comparing the information gain of all the attributes of the dataset ([‘outlook = 0.246’, ‘temp = 0.029’, ‘humidity = 0.152’, ‘windy = 0.048’]), it is observed that ‘outlook’ has the highest information gain, $IG(S, A = 'outlook') = 0.246$. Thus, the first node is selected as 'outlook'.